

# Exponential and logarithmic functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

## Contents

|   |                       |   |
|---|-----------------------|---|
| 1 | Exponential functions | 2 |
| 2 | Logarithmic functions | 2 |
| 3 | Change of base        | 3 |

# 1 Exponential functions

## Definition 1: exponential functions

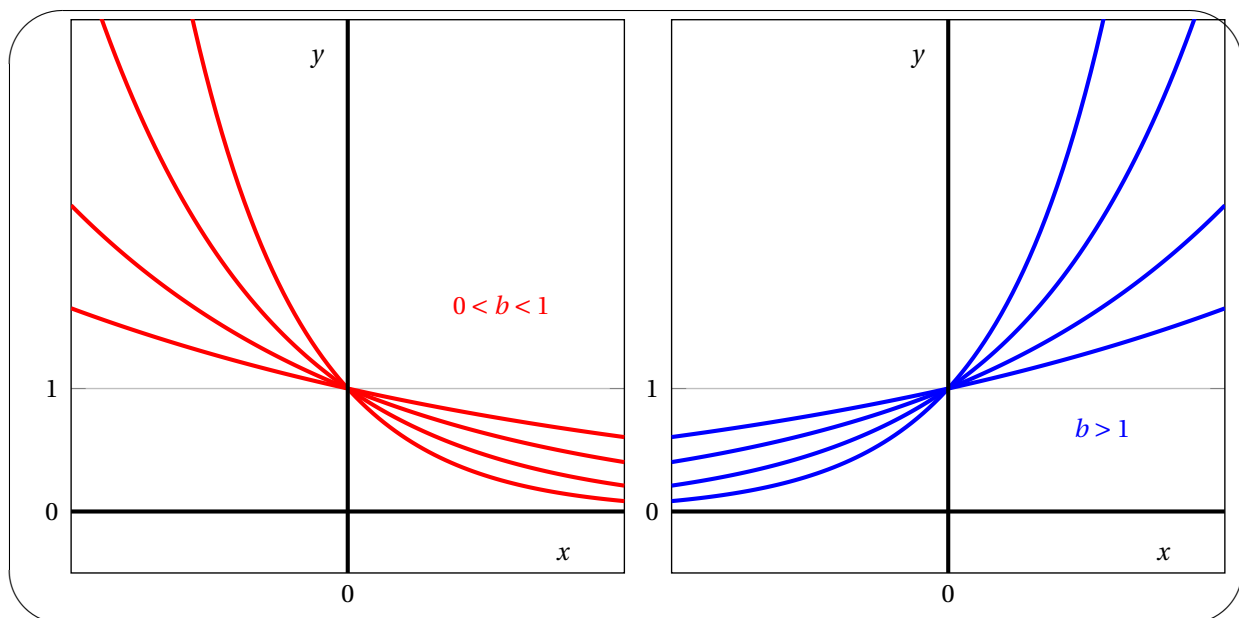
Given  $b \in \mathbb{R}_+ \setminus \{1\}$ , the function:

$$f: \mathbb{R} \rightarrow \mathbb{R}, \quad f: x \mapsto b^x \quad (1)$$

is called the **exponential function** with base  $b$ .

- Regarding positivity and monotonicity, we have:

$$b^x > 0, \quad \forall x \in \mathbb{R} \quad \text{and} \quad \begin{cases} \text{if } 0 < b < 1 \text{ then } f \text{ is decreasing on } \mathbb{R} \\ \text{if } b > 1 \text{ then } f \text{ is increasing on } \mathbb{R} \end{cases}$$



# 2 Logarithmic functions

## Definition 2: logarithmic functions

Given  $a \in \mathbb{R}_{>0} \setminus \{1\}$ , the function:

$$f: (0, +\infty) \rightarrow \mathbb{R}, \quad f: x \mapsto \log_a x \quad (2)$$

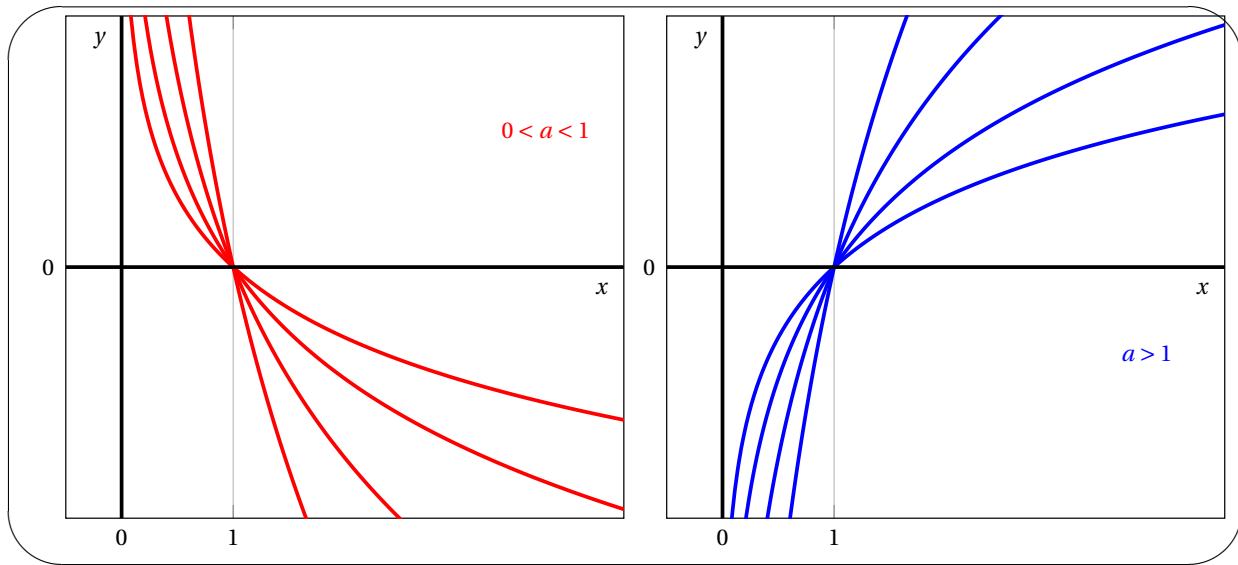
is called the **logarithmic function** with base  $a$ .

- Regarding positivity and monotonicity, we have:

$$\begin{cases} \text{if } 0 < a < 1 & \log_a x > 0, \forall x \in (0, 1), \log_a x < 0, \forall x \in (1, +\infty) \text{ and } f \text{ is decreasing on } \mathbb{R}_{>0} \\ \text{if } a > 1 & \log_a x < 0, \forall x \in (0, 1), \log_a x > 0, \forall x \in (1, +\infty) \text{ and } f \text{ is increasing on } \mathbb{R}_{>0} \end{cases}$$

Moreover, we have:

$$\log_a x = 0 \iff x = 1, \forall a \in \mathbb{R}_{>0} \setminus \{1\}$$



### 3 Change of base

Given any base  $c \in \mathbb{R}_{>0} \setminus \{1\}$ , all exponential and logarithmic functions can be rewritten in terms of another base  $d \in \mathbb{R}_{>0} \setminus \{1\}$  as follows:

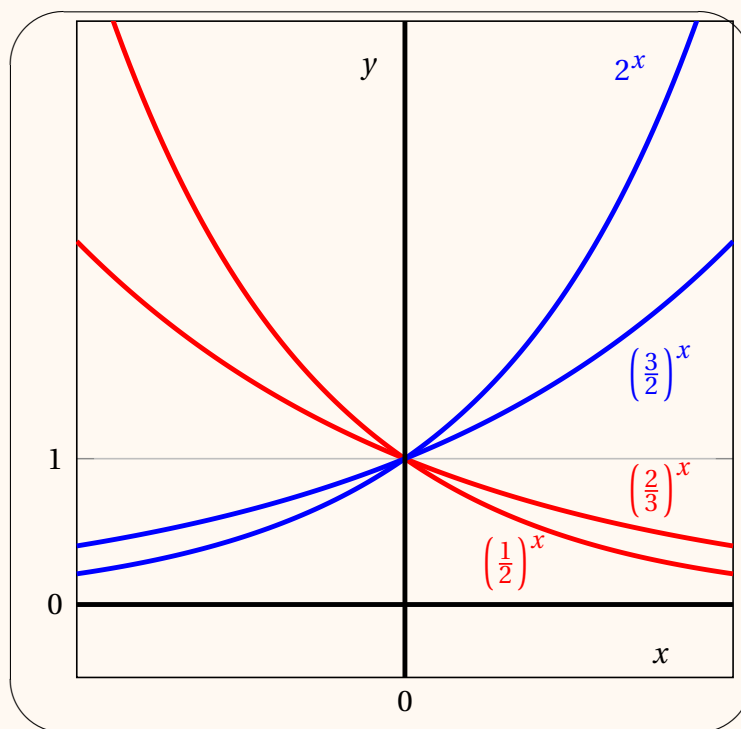
$$c^x = d^{\log_d c^x} = d^{x \log_d c} \quad \text{hence with } \lambda = \log_d c \quad \text{we get} \quad c^x = d^{\lambda x}$$

$$\log_c x = \frac{\log_d x}{\log_d c} \quad \text{hence with } \lambda = \frac{1}{\log_d c} \quad \text{we get} \quad \log_c x = \lambda \log_d x$$

The logarithmic function with base  $e$  is also written  $\log x$  or  $\ln x$ , while the one with base 2 is also written  $\lg x$ .

### Example 1: graphs of exponential and logarithmic functions

Exponential functions:



Logarithmic functions:

